

Title: AI-Powered Banking Customer Complaint Management System using BERT

Abstract

Customer complaints play a significant role in identifying service issues, improving customer satisfaction, and enhancing organizational efficiency. In financial institutions, processing a large number of complaints manually can lead to delays, inconsistent complaint categorization, and inefficient resolution. Traditional complaint management systems require employees to review each complaint, identify its category, assign it to the appropriate department, and determine its priority, making the process time-consuming and prone to human error. Conventional machine learning algorithms such as Logistic Regression, Support Vector Machine (SVM), and Naïve Bayes can perform text classification but often struggle to capture the contextual meaning and semantic relationships present in complex customer complaints.

This project proposes an AI-Powered Customer Complaint Management System using Bidirectional Encoder Representations from Transformers (BERT) to automate complaint classification and management within a banking environment. The proposed system utilizes the Consumer Financial Protection Bureau (CFPB) Consumer Complaint Database (CCDB) to fine-tune a pre-trained BERT model.

The proposed system extends beyond text classification by integrating a complete web-based complaint management platform developed using Flask, MySQL, HTML, CSS, Bootstrap, JavaScript, PyTorch, and the Hugging Face Transformers library. The application consists of three user modules: Customer, Employee, and Administrator. Customers can register, log in, submit complaints, and track the status of their complaint tickets. After the complaint is classified by the BERT model, a business rule engine automatically generates a unique Ticket ID, assigns the complaint to the appropriate department, determines its priority, estimates the response time, and allocates it to the relevant employee based on the predicted complaint category. Employees can view assigned complaints, update their status, add remarks, and resolve complaints, while

administrators can manage users, employees, departments, complaint records, and monitor complaint analytics through an interactive dashboard.

The performance of the BERT-based classifier will be evaluated using Accuracy, Precision, Recall, and F1-score, and compared with traditional machine learning algorithms such as Logistic Regression, Support Vector Machine (SVM), and Naïve Bayes, as well as existing BERT-based implementations. The proposed system aims to reduce manual effort, improve complaint routing accuracy, accelerate complaint resolution, enhance customer experience, and provide an intelligent, automated, and scalable complaint management platform for modern financial institutions.

Datasets

Dataset	URL link
CFPB Consumer Complaint Dataset	https://www.consumerfinance.gov/data-research/consumer-complaints/
Kaggle Automatic Ticket Classification	https://www.kaggle.com/datasets/venkatasubramanian/automatic-ticket-classification
PolyAI-LDN	https://huggingface.co/datasets/PolyAI/banking77

Dataset Description

The proposed system utilizes the Consumer Financial Protection Bureau (CFPB) Consumer Complaint Database (CCDB), which contains real-world consumer complaints related to financial products and services. The dataset includes complaint narratives along with attributes such as Product, Sub-product, Issue, Sub-issue, Company, State, ZIP Code, Date Received, Company Response, Consumer Consent, and Complaint ID.

For this project, only the Consumer Complaint Narrative is used as the input feature for the BERT model, while Product, Sub-product, Issue, and Sub-issue are used as the target labels for complaint classification. These five selected columns were chosen because

they directly contribute to identifying the type and nature of a customer complaint. The remaining attributes, such as Company, State, ZIP Code, Date Received, Company Response, Consumer Consent, and Complaint ID, are excluded from model training because they are either unrelated to complaint classification, unavailable at the time of complaint submission, or may introduce unnecessary bias into the prediction process.

To support the complaint management application, the web system generates additional fields including Customer ID, Customer Name, Department, Assigned Employee, Priority, Ticket ID, Response Time, Complaint Status, Created Date, and Updated Date. These fields are not part of the original CFPB dataset and are generated automatically by the application using predefined business rules after the BERT model predicts the complaint category. The processed dataset is maintained in CSV format for model training and evaluation, while all complaint records and application-generated information are stored in a MySQL database for integration with the web-based complaint management system.

Approach

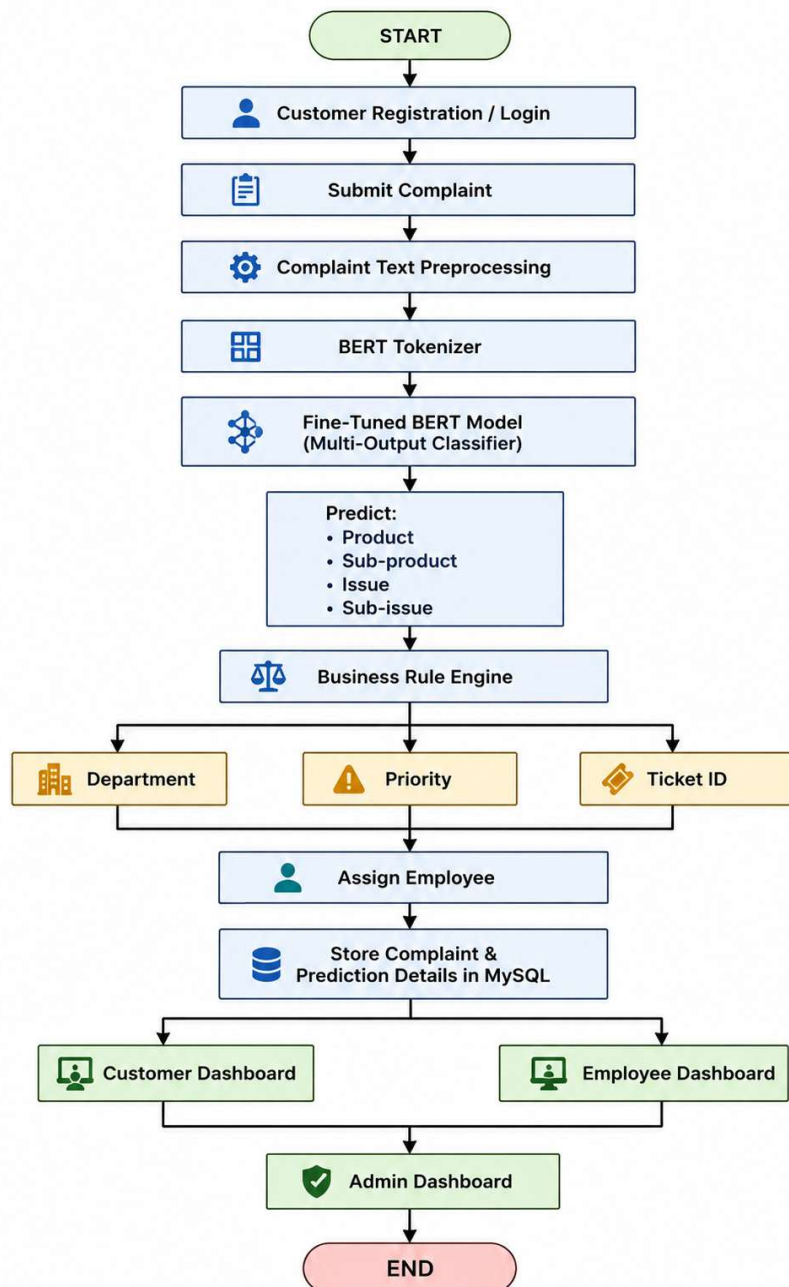
The proposed AI-Powered Customer Complaint Management System consists of three user modules: Customer, Employee, and Administrator.

The workflow begins when a customer registers and logs into the banking complaint portal. The customer submits a complaint by entering the complaint narrative, which is then preprocessed and passed to a fine-tuned BERT model. The model analyzes the complaint text and predicts four complaint attributes: Product, Sub-product, Issue, and Sub-issue.

Based on these predictions, a Business Rule Engine automatically generates a unique Ticket ID, identifies the appropriate Department, determines the complaint Priority, estimates the Response Time, and assigns the complaint to the most suitable employee. All complaint information, including the AI predictions and application-generated details, is stored in a MySQL database.

Employees can log in to view complaints assigned to their department, update complaint status, add remarks, and resolve complaints. Customers can track the status of their complaints through their dashboard using the generated Ticket ID. Administrators can manage customers, employees, departments, complaint records, and monitor system analytics through an interactive dashboard.

The overall system integrates Natural Language Processing (NLP), BERT-based text classification, business rule automation, and web technologies to provide an intelligent, efficient, and scalable complaint management platform.



Project Group Members

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